

EXACT PRIME-COUNT HOMOLOGY OF THE SPHENIC WINDOW COMPLEX

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ABSTRACT. Fix a threshold N and let the window be $W = (N/2, N]$. The *sphenic window complex* K_N is the 2-dimensional simplicial complex whose triangles are the sphenics (squarefree integers pqr with three distinct prime factors) lying in W , whose edges are the prime pairs that co-occur in such a triangle, and whose vertices are the primes involved. These complexes arise in connection with S. Banerjee's question on the multiplicity of the eigenvalue -1 of the integer coprime graph [3]. We reduce the entire homology of K_N to prime-counting functions, with no linear algebra left in the answer, and we determine an exact rational limiting density.

We prove a purely combinatorial identity (Theorem A): for any finite *slab* R , that is any difference of two order ideals in the coordinatewise order on strictly increasing integer tuples, the dimension of the top boundary-cycle space equals the number of *fitting* cells, those x with the all-ones shift $x + (1, \dots, 1)$ still in R , over every field, with an explicit ± 1 integral basis. The sphenic complex is the length-3 slab in the prime-position ruler, so Theorem A gives $\text{rank } \partial_2 = F - \text{elem}$ exactly (the rs-identity), hence $b_1 = (E - F) - V + 1 + \text{elem}$. A finite-parts cancellation collapses this to a signed sum of fourteen prime counts (Theorem B), whose leading behaviour gives an exact rational density $b_1/\pi(N/6) \rightarrow 6H - 3/22 = 97360699/1078282205 \approx 0.0902924$ (Theorem C), unconditional on the prime number theorem, Bertrand's postulate, Dusart's explicit prime-gap bounds, and a finite computation. The top Betti number is $b_2 = \text{elem} = F - D(N/2) + \text{cap}$ for every N , so the full profile is $(b_0, b_1, b_2) = (1, b_1, \text{elem})$ for all $N \geq 154$.

We close with an honest negative. The prime counts of Theorem B carry the non-trivial zeros of ζ , and their amplitudes can be read off the fluctuation of b_1 ; this is exactly the classical explicit-formula duality present in any prime count, and we verify directly that only the prime ruler produces the zeros. We claim **nothing** toward the Riemann hypothesis and no cheaper route to primes or zeros: the construction consumes the primes as input.

1. THE SPHENIC WINDOW COMPLEX, AND BANERJEE'S MOTIVATION

1.1. The object. Fix N . The *window* is $W = (N/2, N]$. A *sphenic* is a squarefree integer with exactly three distinct prime factors, $n = pqr$ with $p < q < r$. We build a 2-complex K_N : the *vertices* are the primes dividing some sphenic in W (to leading order, the primes up to $N/6$); the *triangles* (2-cells) are the sphenics in W , one 2-simplex $\{p, q, r\}$ each; the *edges* are the prime pairs $\{p, q\}$ that co-occur in at least one such triangle. Write V, E, F for the numbers of vertices, edges and triangles, and let ∂_2 be the simplicial boundary $[p, q, r] \mapsto [q, r] - [p, r] + [p, q]$. The two invariants of interest are $\text{rank } \partial_2$ over $\mathbb{Q}$ and the first Betti number $b_1 = E - (V - 1) - \text{rank } \partial_2$.

1.2. Why this complex. Banerjee [3] studies the coprime graph of integers and gives a lower bound on the multiplicity of the eigenvalue -1 , leaving the exact value open. In the author's Papers 1 and 2 [1, 2] that multiplicity is decomposed over the squarefree layers of the integers, turning Banerjee's bound into an equality; the sphenic (three-prime) layer contributes exactly the homological data of K_N , the first Betti number b_1 and a companion count. The present paper, the third in that series, determines this layer: it pins, exactly and rationally, the signed density underneath Banerjee's error term. It does not, on its own, close the full multiplicity,

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which additionally requires the unsigned rank and the other odd layers (see §7 and the companion Paper 4).

1.3. The result in one paragraph. The first Betti number is governed by a completely non-arithmetic fact about staircase-shaped regions of a lattice (§2). Applied to the sphenic window that fact collapses, after an exact cancellation, into a short signed sum of prime counts (§3–§4). Dividing by the vertex count and letting N grow gives an exact rational constant (§5). The top Betti number is a single semiprime count away from the same machinery (§6). Finally (§7) we delimit the tempting appearance that the object “sees” the Riemann zeros: it is the ordinary duality and nothing more.

2. THE COMBINATORIAL CORE: THE SLAB IDENTITY

2.1. Setup. Fix an integer $\omega \geq 2$. Let P_ω be the set of strictly increasing ω -tuples of positive integers, ordered coordinatewise: $x \leq y$ iff $x_t \leq y_t$ for every t . A subset $D \subseteq P_\omega$ is a *down-set* (order ideal) if $y \in D$ and $x \leq y$ imply $x \in D$. A *slab* is a difference $R = D_{\text{hi}} \setminus D_{\text{lo}}$ of two down-sets, $D_{\text{lo}} \subseteq D_{\text{hi}}$; every slab in this paper is finite. Regard each ω -tuple as an $(\omega - 1)$ -simplex on its entries, with the alternating boundary $\partial = \partial_{\omega-1}$; restricting to the cells of R gives a chain group and a boundary map. Write $\delta = (1, \dots, 1)$, call $x \in R$ *fitting* if $x + \delta \in R$, and write $x \ll y$ for strict domination, $x_t < y_t$ for every t .

2.2. The theorem.

Theorem A (slab identity). [PROVED; BANKED AS C-0744] *Fix $\omega \geq 2$ and let $R = D_{\text{hi}} \setminus D_{\text{lo}}$ be a finite slab in P_ω . Then, over every field,*

$$\dim \ker \partial|_R = \#\{x \in R : x + \delta \in R\} = (\text{number of fitting cells}).$$

Moreover $\ker_{\mathbb{Z}} \partial|_R$ is free with an explicit basis of ± 1 vectors (§2.5), and every invariant factor of $\partial|_R$ equals 1, so there is no torsion anywhere in the Smith normal form.

The proof has two ingredients: a domination lemma giving the rank lower bound (the non-fitting boundaries are independent), and an explicit family of cycles giving the matching upper bound (each fitting cell carries a cycle expressible in strictly larger cells). Both are field-agnostic and integral.

2.3. Lemma 1: the Dominated-Pair Lemma.

Lemma 1 (Dominated-Pair Lemma). [PROVED] *In the support of any nonzero $(\omega - 1)$ -cycle z , over any field, there exist cells $x \ll y$. Equivalently, a $\ll$ -antichain of cells has linearly independent boundaries.*

Proof, base $\omega = 2$. A nonzero edge-chain with $\partial z = 0$ gives every vertex of the support graph signed degree zero, hence degree at least 2, so the support contains a graph cycle C . Let a^* be the smallest vertex of C and $(a^*, p), (a^*, q)$ its two edges on C , with $p < q$. Walk C from p to q avoiding a^* ; every vertex on this walk exceeds a^* . Since the walk ends at $q > p$ it contains an edge $g = (c, d)$ (sorted) with $d > p$ and $c > a^*$. Then $(a^*, p) \ll (c, d)$. $\square$

Proof, induction step $\omega \geq 3$ (assuming the lemma at $\omega - 1$). Let $z \neq 0$ with $\partial z = 0$, and let v be the smallest vertex in $\text{supp}(z)$. Split $\text{supp}(z) = A \sqcup B$, where A is the cells containing v (as first coordinate) and B the rest, whose vertices all exceed v . Write each $x \in A$ as $v * x'$ with tail x' an increasing $(\omega - 1)$ -tuple on values above v ; the tails are distinct. From $\partial(v * x') = [x'] - v * \partial[x']$, sort facets by whether they contain v :

- the v -part gives $v * (\sum_A c_x \partial x') = 0$, and $v *$ is injective on v -free chains, so $w := \sum_A c_x [x']$ is a cycle, nonzero since $A \neq \emptyset$, the tails are distinct, and the c_x are nonzero;
- the v -free part gives $\sum_A c_x [x'] + \sum_B c_y \partial y = 0$.

Apply the lemma at $\omega - 1$ to w : there are tails $e \ll f$ in $\text{supp}(w)$. Since the coefficient of $[f]$ in w is nonzero, the v -free relation forces f to appear in ∂y for some $y \in B \cap \text{supp}(z)$, say $y = f \cup \{s\}$ with $s \notin f$, $s > v$.

Insertion claim: $(v, e) \ll y$. The first coordinate satisfies $v < y_1$ since every entry of y exceeds v . Let s land at sorted index k of y , so $y_j = f_j$ for $j < k$, $y_k = s$, $y_j = f_{j-1}$ for $j > k$. For each $t \in \{1, \dots, \omega-1\}$: if $t+1 < k$ then $y_{t+1} = f_{t+1} > f_t > e_t$; if $t+1 = k$ then $y_k = s > f_{k-1} > e_{k-1} = e_t$; if $t+1 > k$ then $y_{t+1} = f_t > e_t$. In every case $e_t < y_{t+1}$, so $(v, e) \ll y$ with both cells in $\text{supp}(z)$. $\square$

The step never uses the slab structure, and a degenerate base re-proves $\omega = 2$, so the whole lemma is one induction from a trivial base.

2.4. Lemma 2: join modes.

Lemma 2 (join modes). [PROVED] *Let x be a fitting cell of the slab R . Partition x into maximal runs of consecutive integers. For a run occupying values $a, a+1, \dots, b$, let its factor be the boundary cycle of the simplex on $\{a, \dots, b+1\}$ (the alternating sum of its subsets each omitting one value). Let J_x be the join of the run factors, with product signs. Then: (1) J_x is a nonzero cycle; (2) $\text{supp}(J_x) = \{y : x \leq y \leq x + \delta\}$ exactly, and every such y lies in R by the sandwich $y \leq x + \delta \in D_{\text{hi}}$ and $y \geq x \notin D_{\text{lo}}$; (3) every coefficient of J_x is ± 1 , and x is the unique coordinatewise minimum of $\text{supp}(J_x)$.*

Consequently, for any strictly monotone ruler (a value function that is a linear extension of the coordinatewise order), every support cell of J_x other than x has strictly larger value, so ∂x lies in the span of $\{\partial y : y \in R, \text{val}(y) > \text{val}(x)\}$. This resolves the apparent collision obstruction: a fitting cell with $x_{t+1} = x_t + 1$ does not carry a full 2^ω box, but the degenerate box is a join of higher simplex boundaries (for one collision, the suspension of a triangle boundary, a 2-sphere on six cells).

2.5. Assembly, and the integral basis. *Non-fitting cells form a $\ll$ -antichain.* If $x, y \in R$ are non-fitting and $x \ll y$, then $x + \delta \leq y \in D_{\text{hi}}$ and $x + \delta \geq x \notin D_{\text{lo}}$ give $x + \delta \in R$, so x was fitting, a contradiction. *rank $\geq \# \text{non-fitting}$:* a nontrivial dependency among the non-fitting boundaries is a nonzero cycle on a $\ll$ -antichain, contradicting Lemma 1. *rank $\leq \# \text{non-fitting}$:* order the cells by strictly decreasing ruler value and eliminate; when a fitting x arrives, Lemma 2 writes ∂x as a ± 1 combination of strictly larger-value boundaries, already processed, so x adds no pivot. Hence $\text{rank} = \# \text{non-fitting}$ and $\dim \ker = \# \text{fitting}$, over every field.

For the integral statement, order the cells by any linear extension of the coordinatewise order. For each fitting x , the vector J_x has its least nonzero entry at row x with value ± 1 (Lemma 2), and distinct fitting cells give distinct pivot rows, so $\{J_x\}$ is in permuted echelon form with unit pivots. This gives independence over every field; saturation of the integer span (eliminate along the ± 1 pivots); and hence $\ker_{\mathbb{Z}} \partial|_R = \mathbb{Z}\text{-span}\{J_x\}$ is free of rank $\# \text{fitting}$. Finally $\text{rank}_{\mathbb{F}_p} \partial = \# \text{non-fitting} = \text{rank}_{\mathbb{Q}} \partial$ for every prime p , so no invariant factor is divisible by any prime and the Smith normal form is $(1, \dots, 1, 0, \dots, 0)$.

2.6. Placement, and what is classical. The strict-domination order is exactly the shifting order $x + \delta \leq y$ of Klivans [5], underlying Kalai's algebraic shifting. *The pure case is Björner–Kalai [4]:* for $D_{\text{lo}} = \emptyset$ the complex is a near-cone, and their theorem gives $\beta_{\omega-1} = \#\{\text{top cells avoiding vertex } 1\}$; the bijection $x \mapsto x - \delta$ carries these onto the fitting cells, so for pure down-sets Theorem A is the classical theorem in one line. *The slab case is the new content, and it is not relative homology:* the relative top cycles of $(D_{\text{hi}}, D_{\text{lo}})$ need boundary only in D_{lo} , whereas our cycles need boundary zero in the ambient complex, a strictly smaller space, so Duval's relative-shifting inequality [6] does not settle it. The nearest machinery is Nagel–Reiner [8]: their Definition 3.4 names exactly our slabs, and their complex-of-boxes cells are the same species as our join modes, but their theorems compute the minimal free resolution of the associated ideal (a different functor); their skew results

are $d = 2$ only, and the general- d skew case is their open Question 5.1. The honest verdict is that Theorem A was not located there; the novelty is unclear pending a specialist, and any external write-up must cite Björner–Kalai (pure case) and Nagel–Reiner (box complexes). A residual check against Duval’s relative Laplacian spectral recursion [7] is the one remaining place the count could already be implicit.

3. THE SPHENIC COMPLEX AS A SLAB, AND THE RS-IDENTITY

Number the primes $p_1 = 2, p_2 = 3, p_3 = 5, \dots$ and give each prime the coordinate equal to its position i . A sphenic $p_i p_j p_k$ becomes the increasing triple (i, j, k) ; its value $p_i p_j p_k$ is used only to test window membership. The all-ones shift $(i, j, k) \mapsto (i + 1, j + 1, k + 1)$ is exactly “advance each prime to the next prime,” $p \mapsto p^+$. Thus the sphenic complex is the $\omega = 3$ slab in the prime-position ruler, and the fitting condition is $p^+ q^+ r^+ \leq N$. Define the *fitting count* $\text{elem}(N) = \#\{\text{sphenic } pqr \in W : p^+ q^+ r^+ \leq N\}$. Theorem A at $\omega = 3$ reads:

$$\boxed{\text{rank } \partial_2 = F - \text{elem}} \quad (\text{the rs-identity}), [\text{PROVED, COROLLARY OF THEOREM A}].$$

Because $\text{rank } \partial_2 = F - \text{elem}$, the first Betti number is a pure count: $b_1 := E - (V - 1) - \text{rank } \partial_2 = (E - F) - V + 1 + \text{elem}$. That it equals the topological first Betti number requires connectivity.

Connectivity lemma. [PROVED; CERTIFIED SELF-CONTAINED WITHIN THE C-0743 BANKING-GRADE PACKET] *For all $N \geq 154$ the complex K_N is connected, so $b_0 = 1$ and the count above is the Betti number b_1 identically.*

The proof is a hub argument through the vertex 2: every prime appearing as a vertex is tied to the 2-hub through window sphenics guaranteed by Bertrand’s postulate, Nagura’s bound, and Dusart’s estimates once $N \geq N_c = 154$, with the component count $c = 1$ checked exactly at four N . The separate edge-support layer-graph lemma (C-0739) is a related but distinct result.

4. AN EXACT PRIME-COUNT FORMULA FOR b_1

4.1. Ingredients. Two facts are used as input; both sit inside the banked C-0743 package (independently re-derived by hand, cleared through four hostile adjudication rounds, with a hashed audit script and a formula-free brute-force confirmation at $N = 1.05, 1.20, 1.50 \times 10^6$).

Master identity. [PROVED; VERIFIED EXACT] *With $B := F - \text{elem}$,*

$$E - B = \pi(N/4) + \text{cap} - D_E - 1,$$

where cap counts the crossing triples $pqr \leq N/2$ with $p^+ q^+ r^+ > N$, and D_E is a blocker/isolation pair count. The transcendental semiprime term $Q_2(N/2)$ cancels identically, which is why no Mertens-type constant survives.

Finite-parts lemma. [PROVED; CERTIFICATE BELOW] *For $N > N_1 = 1,040,255$,*

$$\text{cap} = 10 + \sum_{(p,q) \in \mathcal{T}} [\pi(N/2pq) - \pi(N/p^+ q^+)], \quad D_E = [\pi(N/4) - \pi(N/6)] + \sum_{p \in \{3,5,7\}} [\pi(N/2p p^-) - \pi(N/p p^+)],$$

where $\mathcal{T} = \{(2,3), (2,5), (2,7), (3,5), (3,7), (3,13), (3,19), (3,23), (5,7), (7,13)\}$ is the set of ten crossing pairs.

The threshold N_1 comes from a finite family of at most 601 late-crossing triples, dominated by $(p^+/p)(q^+/q) \leq 1.98925 < 2$ together with Dusart’s estimates, which certify the largest such third prime satisfies $r < 14,872$; the enumerated family’s observed maximum is $r = 5,591$. The certified bound and the observed maximum are distinct numbers, both kept on the record.

4.2. The cancellation. From $b_1 = (E - F) - V + 1 + \text{elem} = (E - B) - (V - 1)$ and the master identity, $b_1 = \pi(N/4) - \pi(N/6) + \text{cap} - D_E$. Insert the finite-parts forms. The leading $\pi(N/4) - \pi(N/6)$ cancels the $p = 2$ term of D_E ; the three cap-pairs $(2, 3), (3, 5), (5, 7)$ cancel term for term against the blocker terms for $p = 3, 5, 7$ (denominators $12/15, 30/35, 70/77$ match exactly). Seven cap-pairs and the constant 10 survive.

Theorem B. [PROVED, GIVEN THE INPUTS ABOVE; VERIFIED] *For all $N > 1,040,255$,*

$$b_1(N) = 10 + \sum_{(p,q) \in \mathcal{P}} [\pi(N/2pq) - \pi(N/p^+q^+)], \quad \mathcal{P} = \{(2, 5), (2, 7), (3, 7), (3, 13), (3, 19), (3, 23), (7, 13)\}.$$

Written out,

$$\begin{aligned} b_1(N) = & 10 + \pi(N/20) - \pi(N/21) + \pi(N/28) - \pi(N/33) + \pi(N/42) - \pi(N/55) \\ & + \pi(N/78) - \pi(N/85) + \pi(N/114) - \pi(N/115) + \pi(N/138) - \pi(N/145) \\ & + \pi(N/182) - \pi(N/187). \end{aligned}$$

The constant 10 counts, for each crossing pair, the single completer dropped on the lower edge of the open interval $\pi(N/2pq) - \pi(N/p^+q^+)$ (the largest prime $\leq N/p^+q^+$, whose successor clears N); these ten dropped completers are genuine cells, [VERIFIED 10/10 AT $N = 2 \times 10^6$ AND 5×10^7].

4.3. Check. The difference $b_1 - (\text{seven-pair sum})$ equals 10 exactly at nine values of N from 1.105×10^6 to 1.20×10^8 , with b_1 computed independently as $(E - F) - V + 1 + \text{elem}$ from the enumerated complex; the seven-pair sum reproduces the entire oscillation of b_1 to numerical identity.

5. THE LIMITING DENSITY

Each bracket satisfies $\pi(N/2pq) - \pi(N/p^+q^+) \sim (N/\log N)(1/2pq - 1/p^+q^+)$, and $V = \pi(N/6) \sim N/(6 \log N)$.

Theorem C. [PROVED, UNCONDITIONAL ON PNT + BERTRAND + DUSART + A FINITE COMPUTATION; BANKED AS C-0743, BANKING-GRADE]

$$b_1(N)/V(N) \longrightarrow K := 6H - \frac{3}{22} = \frac{97360699}{1078282205} = 0.0902924 \dots,$$

$$\text{where } H = \sum_{\text{ten pairs}} (1/2pq - 1/p^+q^+) = \frac{488798363}{12939386460}.$$

The limit is positive: writing $K = 6(\frac{1}{4} - \frac{7}{66} + H) - 1$, positivity is $H > 1/44$, and $1/60 + 1/420 + 5/924$ already exceeds $1/44$. Equivalently, the three cancelled pairs of H sum to exactly $1/44$, so $K = 6 \times (\text{seven-pair sum})$, a sum of seven positive brackets. No transcendental constant appears: the leading (Mertens-type) prime-count constant cancels between the numerator's pieces, which is why K is rational. Here K is the density of the signed first Betti number b_1 ; the lab separately tracks the canonical density $(b_1 - \delta)/V \approx 0.0585$, where $\delta = \text{rank}_{\text{uns}} \partial_2 - \text{rank}_{\text{sgn}} \partial_2$ is a torsion-like frustration term, not known to be rational and where any remaining transcendence lives (the companion Paper 4 reduces δ to counting at a certified perimeter). These two numbers must not be confused.

6. THE TOP BETTI NUMBER, AND THE FULL PROFILE

The complex has no 3-cells, so $b_2 = \dim \ker \partial_2 = \text{elem}$ by Theorem A directly. What C-0745 adds is a closed form for elem through the advance map.

Advance bijection. [PROVED; BANKED AS C-0745] *The next-prime map $(p, q, r) \mapsto (p^+, q^+, r^+)$ carries increasing prime triples bijectively onto increasing odd-prime triples (inverse: the previous-prime map, one-line surjectivity). The shell count*

$$J(N) := \#\{pqr \leq N : p^+q^+r^+ > N\} = D(N/2) = \#\{\text{odd semiprimes} \leq N/2\} \quad \text{exactly, every } N.$$

The lower-bound step is one line: $p^+q^+r^+ > pqr > N/2$, so a shell triple's image is an odd semiprime $\leq N/2$, and the bijection matches the counts (recovering 168330 at $N = 2 \times 10^6$). Splitting at the window edge gives, with no threshold,

$$\text{rank } \partial_2 = D(N/2) - \text{cap}, \quad b_2 = \text{elem} = F - D(N/2) + \text{cap} \quad \text{exact for every } N.$$

Above the threshold this inherits the ten-pair π -form of cap: for $N > 1,040,255$,

$$b_2(N) = S(N) - S(N/2) - D(N/2) + 10 + \sum_{\text{ten pairs}} [\pi(N/2pq) - \pi(N/p^+q^+)],$$

with S the sphenic count, $F = S(N) - S(N/2)$. Verified exact at $N \in \{2 \times 10^5, 5 \times 10^5, 1.05 \times 10^6, 2 \times 10^6, 5 \times 10^6, 2 \times 10^7\}$; below N_1 the π -form is off by exactly 2.

Full profile. [PROVED] *For all $N \geq 154$, $(b_0, b_1, b_2) = (1, b_1, \text{elem})$, with $b_1 = 10 + \text{fourteen prime counts for } N > 1,040,255$, $b_1/V \rightarrow 6H - 3/22$, and $b_2 = \text{elem} = F - D(N/2) + \text{cap}$. No linear algebra survives in any of the three numbers.*

For Banerjee's problem this supplies the three-prime layer in closed form: the signed density his error term skirts is pinned exactly and rationally. The full -1 multiplicity still needs the unsigned rank (δ , addressed at the certified perimeter in the companion Paper 4) and the other odd layers.

7. THE SPECTRAL SHADOW, AND EXACTLY HOW FAR IT GOES

7.1. What is true. Because b_1 is a fixed signed combination of prime counts (Theorem B), its fluctuation about the smooth part is, by the classical explicit formula, a sum over the non-trivial zeros ρ of ζ :

$$b_1(N) - (\text{smooth}) = - \sum_{\rho} M(\rho) \cdot (N^{\rho}\text{-type term}), \quad M(\rho) = \sum_m \pm m^{-\rho},$$

the sum over the fourteen denominators m . The factor $M(\rho)$, a Dirichlet polynomial in fourteen small integers, is derived with no reference to ζ . Scored against the measured per-zero amplitudes of b_1/V (data to $N = 6 \times 10^{10}$, the known zeros used only as sampling frequencies), it predicts the relative amplitudes with correlation 0.992 over the first 30 zeros (0.974 without the dominant first zero), matching into the noise floor (γ_{24} measured 0.026 vs. predicted 0.027; γ_{30} 0.026 vs. 0.028), [VERIFIED]. Reading zeros directly off the spectrum recovers 11 of them, up to the 18th near $\gamma_{18} \approx 72.1$, resolution-limited not signal-limited (the count grows 6, 9, 11 as sieve depth grows $3 \times 10^8, 6 \times 10^9, 6 \times 10^{10}$). The fluctuation grows no faster than $N^{1/2}$ (measured exponent 0.44), consistent with and required by the Riemann hypothesis; the fourteen-term combination cancels about 99.6% of the individual prime-count fluctuation, so b_1 is an unusually quiet object.

The dartboard null [VERIFIED]: tested against 20,000 loose controls (fourteen random distinct integers in $[10, 200]$, balanced signs) and 20,000 matched controls (seven crossing-style pairs with ratio in $[1.005, 1.6]$), the true envelope scores 0.998 (0.993 excluding γ_1) against a null median 0.64 and null maximum 0.94 across all 40,000 draws. With the shared decay stripped, the null collapses to median ≈ 0.00 while the true envelope holds 0.992. The amplitude channel is specific to the topology-selected denominators, not an artifact of shared decay.

7.2. What this is not. All of the above is the ordinary prime-to-zero duality; the zeros are present because b_1 is built from prime counts, and every prime count contains them (Riemann, 1859). Two controls make the point. *The envelope is blind to the zeros' positions:* $|M(\frac{1}{2} + it)|$ averages 0.489 at the 30 true zeros and 0.500 at 5,000 random heights, so the zeros sit at the 48th percentile of M 's own values. *Only the prime ruler throws them:* the identical construction with the plain-integer ruler peaks at the window-beat frequency, not the first zero, while the prime ruler puts the first zero far above everything else. Consequently we claim [NOT CLAIMED] any progress on the Riemann

hypothesis, any new bridge between primes and zeros, or any cheaper route to either: computing b_1 requires the primes as input, and nothing is obtained more cheaply than it is supplied.

8. DISCUSSION

The self-contained core is the rational density $K = 6H - 3/22$ (Theorem C) and its exact finite- N form as fourteen prime counts (Theorem B), resting only on PNT, Bertrand, Dusart, and a finite certificate; this pins the signed density Banerjee’s error term skirts, and the full profile supplies the whole three-prime layer as counting functions. The most broadly interesting piece may be Theorem A itself, a statement with no arithmetic in it: its pure-down-set case is classical (Björner–Kalai), and the slab case is a concrete theorem in a well-studied corner of algebraic combinatorics; whether it is already on record is a question a specialist could settle on sight, and the honest current label is “adjacent, apparently unstated.” What is honestly not here is any route to the Riemann hypothesis or any predictor for primes.

APPENDIX A. THE FOURTEEN DENOMINATORS

Signed, in order: $+20, -21, +28, -33, +42, -55, +78, -85, +114, -115, +138, -145, +182, -187$. Each pair $(2pq, p^+q^+)$ is a crossing pair with $p^+q^+ > 2pq$, so each bracket counts primes in a thin interval and is non-negative.

APPENDIX B. CERTIFICATES AND DATA (REPRODUCIBLE)

All entry points run from the repository root; the repository (concept DOI 10.5281/zenodo.21135221) holds the scripts and the certificate bundles under `certificates/` as gzipped JSON with SHA-256 manifests.

- *Slab identity and field-independence.* $\dim \ker$ over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_7, \mathbb{F}_{101}, \mathbb{F}_{2^{31}-1}$ agree (`slab.gf2_sweep.py`, 149 slabs, 0 mismatches); verified $\omega = 2, 3, 4$ across many rulers, $\omega = 5, 7$ for the prime case; proof verification `dp_proof_verify.py` clean.
- *Theorem B.* b_1 –seven-pair sum = 10 at $N \in \{1.105, 1.986, 3.569, 6.413, 11.52, 20.71, 37.21, 66.87, 120.2\} \times 10^6$.
- *Theorem C finite parts.* $H = 488798363/12939386460$; the D_E constant $7/66 = 1/12 + 1/60 + 1/210 + 1/770$ with blockers $\{2, 3, 5, 7\}$; $K = 97360699/1078282205$; late-crossing family ≤ 601 triples; certified $r < 14,872$ (Dusart) vs. observed maximum $r = 5,591$; hashed audit script reproduces all certificates plus a formula-free brute-force check at $N = 1.05, 1.20, 1.50 \times 10^6$.
- *Top Betti number.* `elem_closed_form_verify.py`: $b_2 = \text{elem} = F - D(N/2) + \text{cap}$ exact at $N \in \{2 \times 10^5, 5 \times 10^5, 1.05 \times 10^6, 2 \times 10^6, 5 \times 10^6, 2 \times 10^7\}$; $J(N) = D(N/2)$ recovers 168330 at $N = 2 \times 10^6$.
- *Spectral shadow.* 30-zero envelope correlation 0.992 (0.974 without the first zero); dartboard null over 40,000 controls (`envelope_dartboard.py`); position-blindness control 0.489 vs. 0.500; growth exponent 0.44.

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